Patrick Thangarajah

INJURY BIOMECHANICS | OCCUPANT SAFETY | CRASH & IMPACT SIMULATION

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Summary

Mechanical Engineering MASc graduate specializing in injury biomechanics, with experience evaluating injury mechanisms through finite element simulation, physical test validation, injury criteria, and peer-reviewed scientific literature.

Skills

Technical Communication Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs

Testing & Analysis DOE, JMP, Minitab, MATLAB, Microsoft Excel, Troubleshooting

Injury Biomechanics HIC15, Nij, chest deflection, AIS injury risk, injury-criteria interpretation, THOR-5F, Hybrid III, GHBMC

Controls & Instrumentation Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice

Education

Master of Applied Science in Mechanical and Mechatronics Engineering *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Jul 2026*

- Research in occupant safety and biomechanics

Bachelor of Applied Science in Mechanical Engineering, Co-op Program *Vancouver, BC, Canada*

University of British Columbia *Sep 2018 - May 2023*

- Biomechanics and Medical Devices Option with coursework in human anatomy, physiology, biomaterials, and biomedical design

Work/Research Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Jul 2026*

- Validated a THOR-5F ATD FE model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Designed and analyzed a 104-simulation FE parametric study of seat belt anchor locations using JMP DOE, least-squares regression, and ANOVA; identified a rearward and inboard position with 16% lower peak chest deflection than the study average
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

Summer Research Student *Vancouver, BC, Canada (Remote)*

Centre for Heart Lung Innovation *May 2022 - Aug 2022*

- Built a Jupyter Notebook transfer-learning script to diagnose diabetic kidney disease using an emerging imaging platform
- Improved diagnostic accuracy by 10% through model optimization, training refinement, and comparison with existing tissue-slide images

R&D Engineer (Part-time) *Remote*

Echoscope *May 2023 - Sep 2023*

- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functional testing; all five were approved and used by the client
- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks

Engineering Intern *Victoria, BC, Canada*

Breathe Global *Jul 2021 - Aug 2021*

- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

Projects

Soft Distensible Bladder Model - Capstone Project *Vancouver, BC, Canada*

University of British Columbia *Sep 2022 - Apr 2023*

- Prototyped bladder phantoms using silicone and agar, comparing pressure-volume response with literature to guide material selection
- Built a control interface for the soft bladder phantom, enabling real-time pressure/volume data logging, PV/PT curve generation, and automated filling/voiding cycles via four adjustable parameters (max volume, flow rate, jet duration, jet interval)
- Validated pressure chamber performance using the integrated sensor and control system, demonstrating stable, repeatable pressurization at 2.2 kPa (13.8s ramp-up) up to a maximum internal bladder pressure of 4.75 kPa

Simulation of Cervical Fractures in American Football (ME 720 Course Project) *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Dec 2023*

- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.